

Static Timing Analysis

The Physics of Digital Time

Structured Learning Notes

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Contents

1 The Core Problem: Physics vs. Logic

Forget software tools for a moment. Think physics. A digital circuit is essentially a race. Signals leave one register, travel through logic gates and wires, and must arrive at the next register at the exact right moment.

- **Too Late:** Setup failure → The wrong value is captured.
- **Too Early:** Hold failure → The new value overwrites the old one too fast.

STA's One Question

“For every possible path in this design, under worst-case conditions, will data arrive at the right time?”

2 Why Simulation Fails (And STA Exists)

2.1 The Simulation Mindset

Simulation is like watching a movie. You choose input vectors, watch what happens, and hope you covered all cases.

- **Problems:** It misses corner cases, it is non-exhaustive, and it is computationally impossible for million-path designs.

2.2 The STA Mindset

STA is like a mathematical proof.

- No input vectors required.
- No guessing.
- **Topological Propagation:** It pushes timing through gates, accumulates delay, and stops at endpoints.

3 Vector-less Analysis: The Mechanism

STA does not care about logical values (“0” or “1”). It ignores functional behavior and focuses entirely on **delay math**.

Pipes: Gates + Wires

Flow Time: Delay

Reservoirs: Registers

4 Timing Goals & Frequency

Every synchronous chip lives under one law: **Everything must fit inside one clock period.**

If your clock is 1 GHz ($T_{period} = 1\text{ns}$):

- All logic between registers must finish in $< 1\text{ns}$ (minus uncertainty margins).
- STA verifies frequency targets, signal collisions, and data integrity.

5 Variability: Why Delay isn't a Single Number

Delays change constantly due to physics. STA uses **PVT** (Process, Voltage, Temperature) variations.

1. **Process:** Transistors vary during manufacturing.
2. **Voltage:** IR drop slows down switching.
3. **Temperature:** Hot silicon is generally slower.
4. **Noise:** Crosstalk and jitter.

Min/Max Analysis

STA must always assume:

- **Worst Case Slow:** For Setup checks (Late Analysis).
- **Worst Case Fast:** For Hold checks (Early Analysis).

6 Registers: The Traffic Lights

Registers define the timing windows. They are not just storage; they are constraints.

Setup Constraint (Max Delay)

Data must arrive **before** the clock edge.

$$T_{data_delay} + T_{setup} < T_{clock_period}$$

Violation: Data arrives too late.

Hold Constraint (Min Delay)

Data must stay stable **after** the clock edge.

$$T_{data_delay} > T_{hold}$$

Violation: Data changes too early (racing through).

7 The Anatomy of a Timing Path

STA enumerates every possible path between valid **Startpoints** and **Endpoints**.

Startpoints	Endpoints
Input Ports	Output Ports
Clock pins of Flip-Flops	Data pins of Flip-Flops

8 Special Path Types

8.1 False Paths

Some paths exist physically but not logically (e.g., mutually exclusive MUX inputs). STA doesn't know logic intent. You must constrain these as `set_false_path`, otherwise STA will report violations for things that never happen.

8.2 Multi-Cycle Paths (MCP)

Logic intentionally designed to take > 1 cycle (e.g., complex arithmetic). By default, STA assumes everything is single-cycle. MCP constraints relax the **Required Time**.

9 The Mathematics of Timing

9.1 Required Time vs. Arrival Time

- **Required Time:** The deadline. “By when *must* data arrive?”
- **Arrival Time:** The reality. “When *does* data actually get there?”

9.2 Slack

Slack is the single most important number in a timing report. It represents your **Margin**.

The Slack Equation

$$\text{Slack} = \text{Required Time} - \text{Arrival Time}$$

- **Positive Slack:** Safe.
- **Zero Slack:** Razor’s edge (exact met timing).
- **Negative Slack:** **FAILURE**.

The path with the most negative slack is your **Critical Path**.

10 Early vs. Late Analysis (Crucial)

Two Universes of Analysis

- **Late Analysis (Setup):** Assumes everything is **slow**. Checks if the signal arrives *before* the next clock edge.
- **Early Analysis (Hold):** Assumes everything is **fast**. Checks if the signal stays stable *long enough* after the current clock edge.

11 Summary: The Mental Model

Think of Static Timing Analysis as a physics-based scheduling problem with strict deadlines (Required Time) and actual physical estimates (Arrival Time).

- STA is not pessimistic by accident; it is pessimistic by **design**.
- If you pass STA, you are mathematically guaranteed to work on silicon (within the modeled physics).